

Personal reflection

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1. Introduction

In this project, I am assigned to work on a web application that can automate some manual work for **the Heineken brewing company**. The task is to create a feature that can automatically look into the company's internal Jira ticket and team meetings and create insight from them that can later convert into meaningful information in the form of an audio podcast. Not only that, but it should also be able to update the internal Jira ticket based on the task completion or the task that still needs to be done.

1.1 My role

My role is to be a connection bridge between the frontend and backend and integrate them along with the first feature that generates audio insight about the company's current affairs.

1.2 The goal

The goal is to ship a working MVP along with maintaining a healthy relationship with the team that includes maintaining meaningful conversation that could lead to maximum success.

2. Situation / Context (What happened?)

2.1 The project

Heineken is the world's largest brewing company that brews many types of alcoholic drinks, including but not limited to beers. Up until now, they have been managing their internal Jira tickets and sources manually; that works well. But it takes quite a valuable amount of time and resources that they could have used for some more important tasks. So to be more efficient in terms of saving these valuable resources, they felt a need to create a system that can automatically decide what to do and let its target audience know about it.

There are **five** features that need to be in mind for this project:

1. Audio briefing for the morning

For the audio briefing that I am originally assigned to, the program supposed to generate a 3 - 4 min audio briefing that talks about the current affairs based on the KPI that the system gets from the internal database. The data that should be converted into an audio briefing can be found from many sources: PDFs, PPT, websites, etc.

2. Meeting action taker

This feature is going to take a summary from video team meetings and Jira tickets and decide how much work needs to be done for the given day. It will automatically close the ticket if it finds a given task is already finished.

The feature will also send notifications to the given users if anything new comes that needs special focus.

3. Shipment SLA

In this feature the system will track the status of the shipment. Since there is no tracking system available or given by the company, we need to rely on the internal Jira environment that is responsible for tracking the status of the shipment vehicle.

Later on, it will create an audio briefing that will describe what's going on with the shipment.

4. Approval workflow

This feature is entitled to decide whether a particular task is finished; if not, it will send an urgent email to the person who is assigned to that task.

5. Evening wrap

This feature will summarize the whole day's work process and see if there is any backlog that is not done yet. Also, it will check tasks based on priorities. If the tasks are too urgent, then it will let the user know about it.

This particular feature concerns the user role, and based on the user's role, it will send customized notifications.

2.2 Timeline / team setup

We were given more or less 3 months to finish this project. We already have worked for 2 months.

There were 5 team members originally, but later on one member didn't show up, so we are taking over their work too.

Our current team strength is 4 people working on 4 different features. But we are also trying to contribute to the fifth feature due to the lack of team members.

3. What I am doing

I am working on feature 1 as mentioned earlier. On top of that, I am building the backend with the help of another team member that can later be integrated to the frontend

we have already created so far. The frontend still needs quite a significant amount of work, so that would be my next priority.

The tools' decision was a collective decision where my influence was a driving force. But I usually try not to dictate the process since it is supposed to be done in a democratic way.

We are using the below tools for this project, but it can be updated later on:

Backend: fastapi (python)

Frontend: react js

Storage: Relational database(sqlite), firebase storage bucket, firebase no sql database

Others: firebase fcm, firebase authentication, ElevenLabs (for audio generation), huggingface pretrained models, langgraph/langchain for a multi-layered agent model (still in process), docker (to make it a single project that can run everywhere any time without the burden of needing to set up for a hour long process.)

4. What Went Well

The overall conversation with the team was fairly smooth, but it could have been better considering only a few people replied instantly. All the team members are maintaining a friendly yet professional relationship, which I appreciate the most.

Even a small concern will drive them to work on it. To me it's a sign of relief. Up until now, we have 4 features up and running that need to be connected to the end project, which is currently under process. But to us it's an iterative process, so further work might be needed after we finish our MVP. But in my opinion that's quite expected.

5. What Went Wrong / Challenges

5.1 Team problems

The biggest hurdle was to find a time slot that would be flexible for 4 of us since each of the team members needs to do part-time jobs in addition to studying.

So far there has been no visible crack between any of our team members. But things get quite dramatic, as I know from my previous experience. But the bright side is that we haven't reached that point yet.

Despite all of my optimism, which I believe is quite accurate, I did feel like some members are not happy with certain other members due to their overpromising or scheduling issues. But I tried to resolve the issue by personally talking with them. But I can not guarantee that this kind of issue won't happen again.

5.2 My personal mistakes

It took me quite a while to get a complete grasp of the project due to lack of information. On top of that, I wasn't quite sure where the situation was going. I still believe some team members still don't have that much of an idea of what they are meant to do.

These minor things delayed quite a significant amount of time before we actually became serious about the project.

But I am in no position to blame anyone since I have my fair share of mistakes that I could have prevented.

I was going through some personal circumstances that made me virtually unable to focus on this project in the initial phase. It took me quite a while to hold my ground again and make myself function again. That process made me constantly procrastinate, which I later managed and reduced quite significantly.

5.3 Technical issues

Initially, we were facing some limitations in terms of using some services (pre-trained models) that were not for free. But we managed to gain premium access by paying some small amount of money that can be neglected.

But there are some important concerns regarding our system that are worth mentioning.

Different team members are using slightly different databases (not everyone but people who needs to rely on databases for their features.)

So when the final version needs to be integrated, we might need to take some other decision, or the synchronization process will be hampered.

6. What I Would Do Differently

Since the lesson has been learned, my next step would be to be as disciplined as possible. The planning that we have done so far should be enough until we finish the project.

But I would definitely make sure that all the team members are communicating properly. During my working on this project, I felt a need to have a team member who could coordinate the tasks and distribute them to team members based on their strengths. So I will make sure that one of us takes that responsibility.

7. Conclusion

This project is quite different from any of my previous projects in this university. It gave me a lot of mixed feelings. Some of them were quite positive; some of them were not so positive, as I have mentioned earlier. Due to being a perfectionist, I personally cannot be completely satisfied with the current progression we have. To me, the time could have been used in a much more creative way. But I also understand that a sheer amount of time is needed to create a proper relationship with the team members. So I believe that, it was an opportunity cost that we all paid.

Overall, I believe every team member has learned from their mistake and is now on the right path towards the end goal. Since we already have quite a significant amount of work done, we can work on the rest of the project with ease.